

Functional Connectivity

How to study the connectivity in the brain with resting state fMRI

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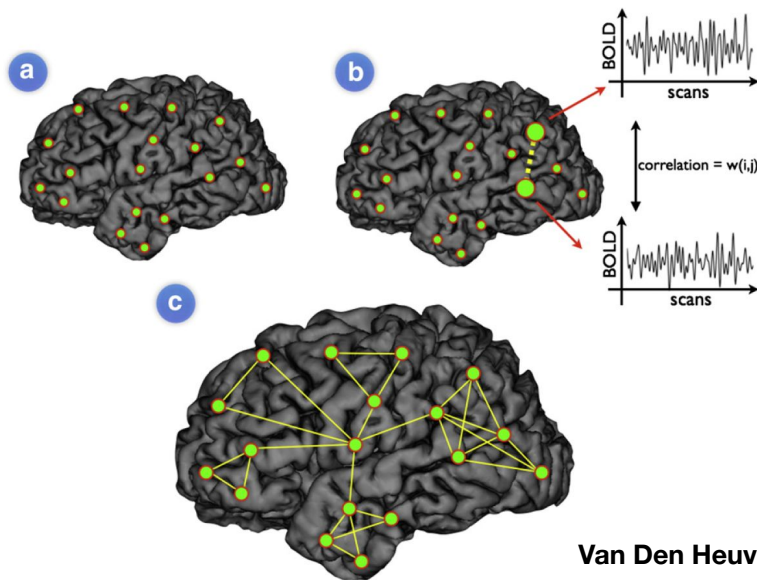
Computational application
to cognitive neuroscience



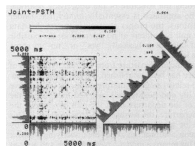
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What does Functional Connectivity (FC) mean?

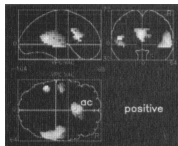
correlation of some kind of neurophysiological signal measured in different brain regions



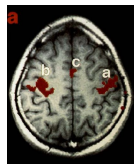
A brief timeline of studies about functional connectivity



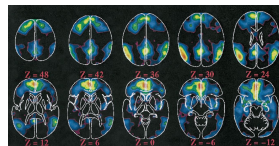
Correlation of spike trains



PCA-based FC with PET

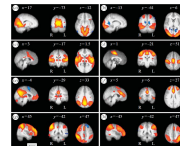


Seed-based FC with rs-fMRI between motor cortices

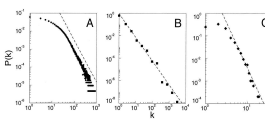
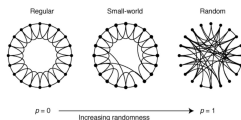
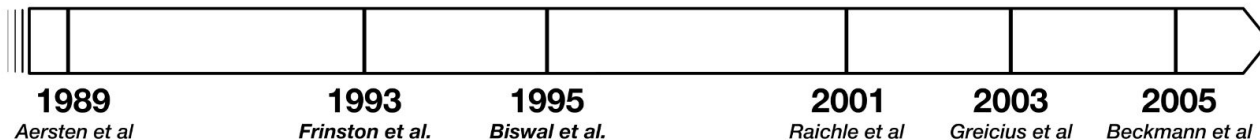


Default-mode network in PET data

Default-mode network in fMRI data



Resting-state networks



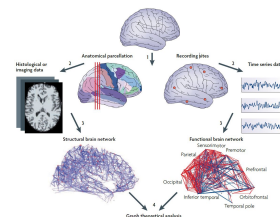
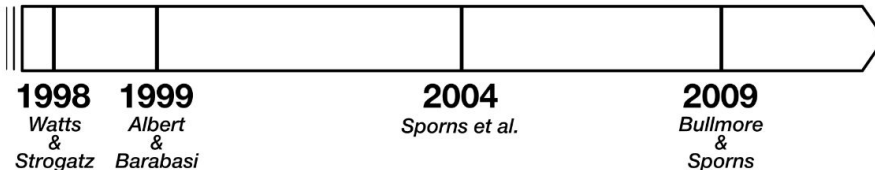
Collective dynamics of small-world networks

Emergence of scaling in random networks

Organization of complex brain networks

Graph-based analysis of complex brain networks

Network Theory



Different approaches to FC with fMRI

model-dependent
(seed methods)

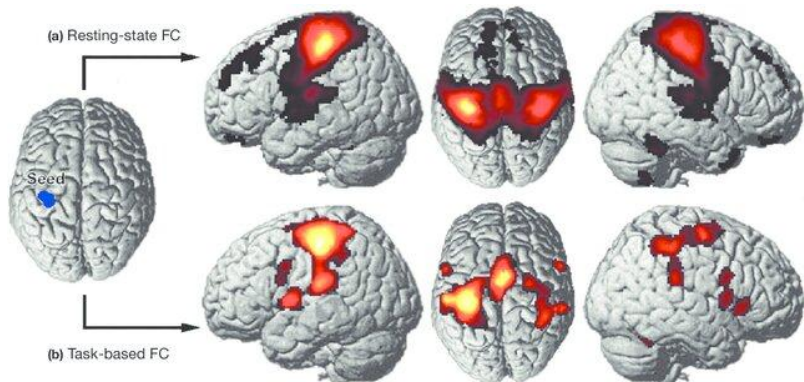
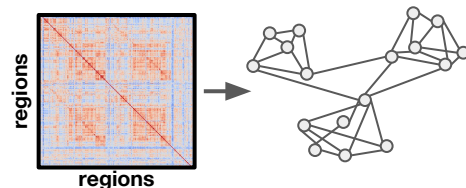
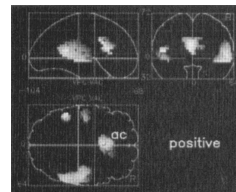
model-free
(without a priori information)

PCA

ICA

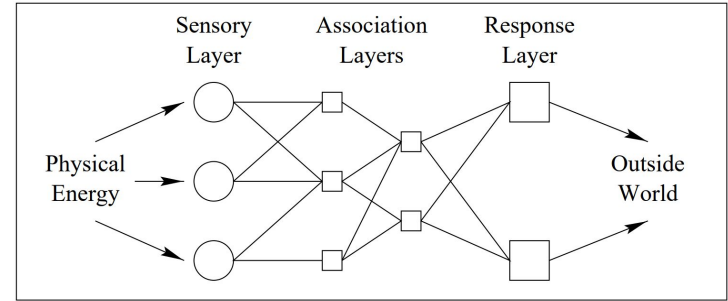
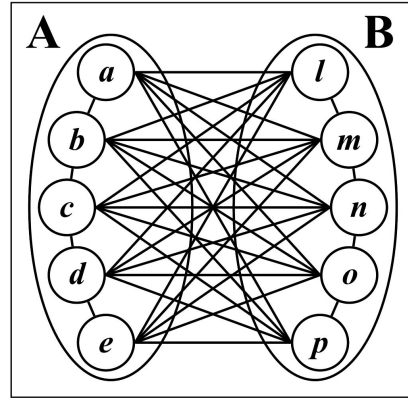
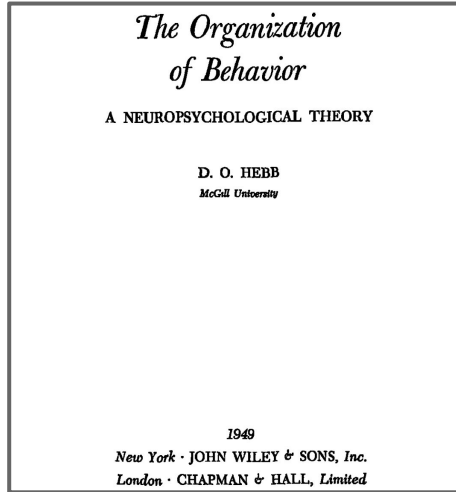
**clustering
methods**

**network-based
analysis**



The idea of a connected brain

has a very long story...

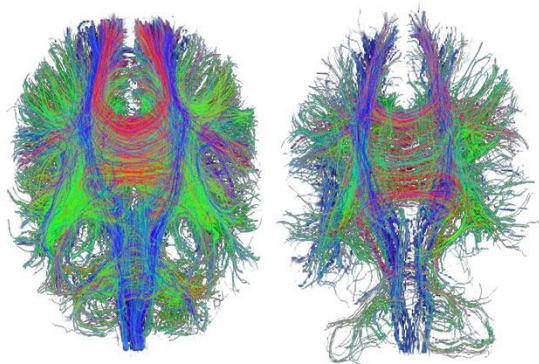


Medler, 1998

...but it was only in the last 3 decades that we managed to observe and analyse the connections inside the brain

Many types of connectivity

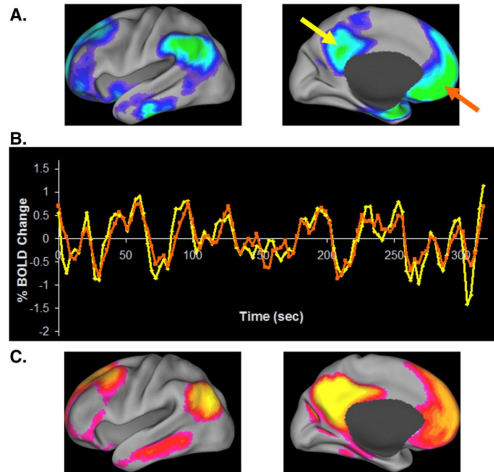
Anatomical



<https://www.umimri.com/dti/>

connections between neurons are too “slow” to explain cognitive functions...

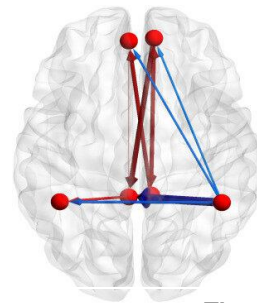
Functional



Raichle, 2007

To study the functional connectivity allows to look at the emergent properties of the brain

Effective



Ting, 2014

Models of causal (and directed) relations between brain regions or group of neurons

Granger causality

Whole brain computational modeling (e.g. Hopf model)

Why the resting-state?

Lowe, 2010

The incredible amount of information in the activity at rest of the brain was discovered studying the “physiological noise” of the brain (cardiac and respiratory related signal).

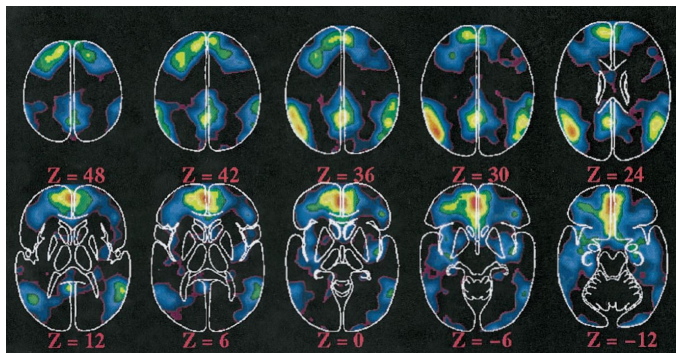
Even removing this kind of noise of the brain, some residual spatio-temporal correlation of the signal remains...

let's try to look together at this correlation between brain regions in the resting-state brain signal

The Default-Mode Network

Since the first experimental observation of the brain in the early 20th century researchers has observed that the amount of activity of the brain at rest is nearly the same as during an active task.

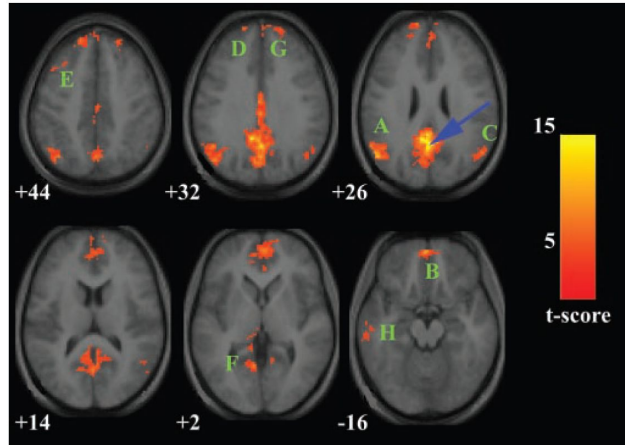
In 1997, Shulman et al. observed the existence of *task-negative network* that seemed to be deactivate during active tasks, like language, motor or visual goal-directed tasks.



The Default-Mode Network

In 2001, Raichle et al. observed the same pattern of activated brain areas in subjects at rest using PET imaging. They coined the term *Default Mode Network* for this particular network.

In 2003, Greicius et al. discover the same network with fMRI resting state data.



The Default-Mode Network

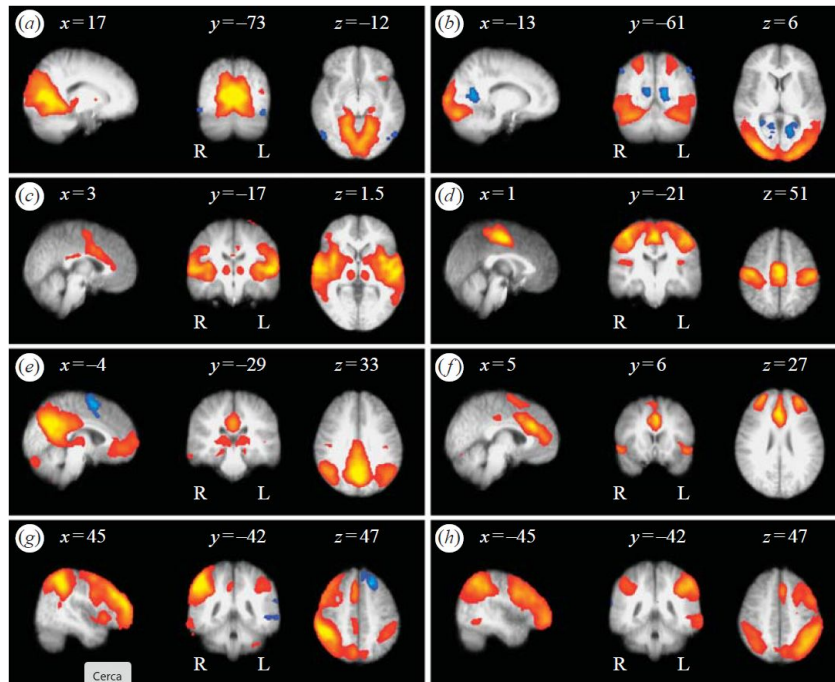
Following this initial works, many studies have addressed the role of the Default Mode Network with respect to representating of the self, thinking about others and remembering the past.

Alteration of the DMN have also been related to clinical disorders such as Alzheimer's disease, autism depression and many others.

The brain region composing the DMN are the medial prefrontal cortex, the posterior cingulate cortex, the precuneus and the angular gyrus.

The Resting State Networks

In 2005, Beckmann et al. Defined a set of resting-state networks performing an ICA analysis over resting-state fMRI data.



Our lectures about functional connectivity

01 - Seed-based FC

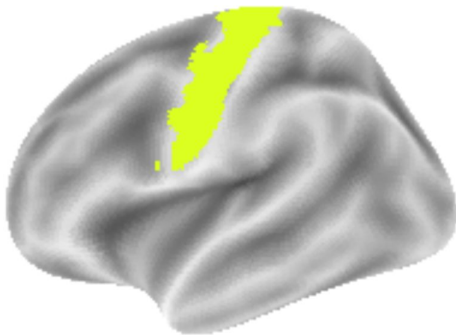
02 - FC with PCA and ICA

03 - Graph analysis applied to resting-state fMRI

04 - Dynamic Functional Connectivity

01 - Seed-based FC

Just choose a seed region of the brain



And then correlate the average time series of the seed with all the voxels of the brain

It's simple :)

02 - PCA and ICA-based FC

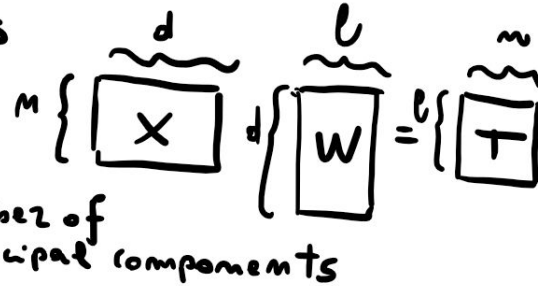
PCA Useful to reduce the dimensionality of high dimensional data

$$XW = T \quad \text{PCA it's a linear transformation}$$

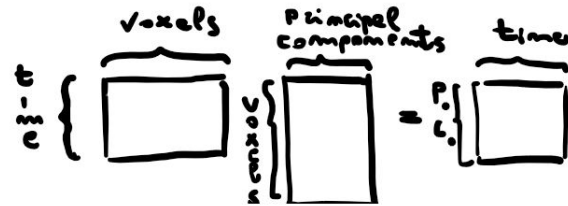
X : $m \times d$ DATA MATRIX m : samples
 d : dimensions

W : $d \times l$ TRANSFORMATION MATRIX

T : $l \times m$ TRANSFORMED DATA



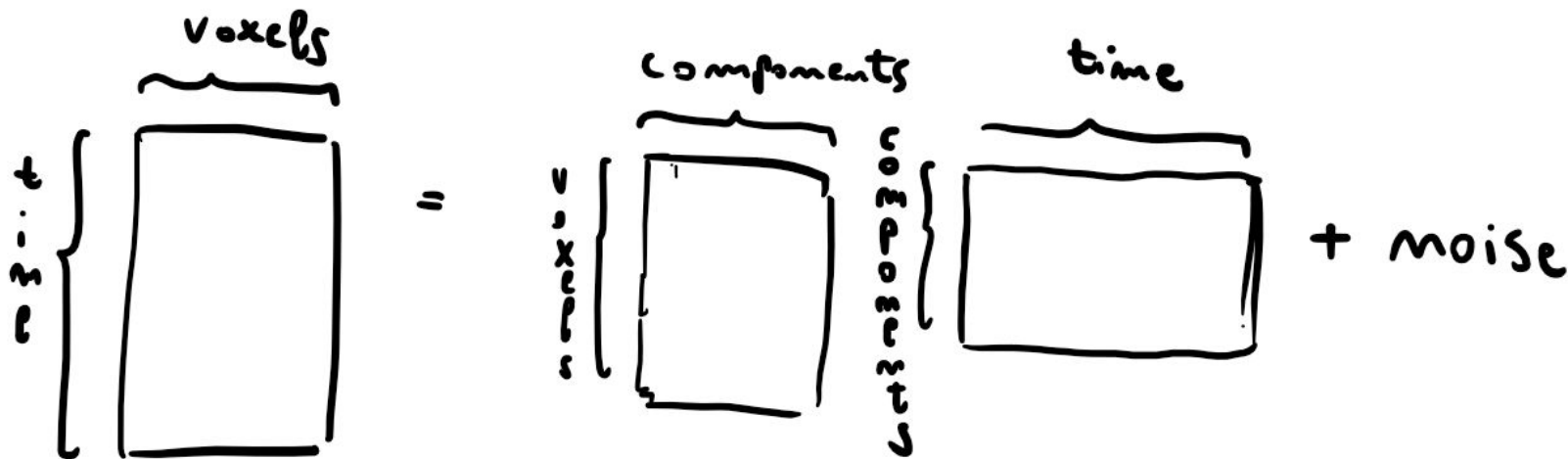
$$W_{(n)} = \underset{\|W\|=1}{\operatorname{argmax}} \left\{ \|\hat{X}_k W\|^2 \right\}$$



The components obtained through the PCA are orthogonal one to each other

02 - PCA and ICA-based FC

ICA Same idea as PCA, but with non-orthogonal components. The new components are statistically independent. With ICA it's also possible to add a noise term.



This is the spatial ICA, but also the temporal ICA exists

03 - Graph analysis applied to resting-state fMRI

Definitions

$G = (V, E) : \textit{Graph}$

$V = \{v_1, \dots, v_n\} : \textit{Set of vertices}$

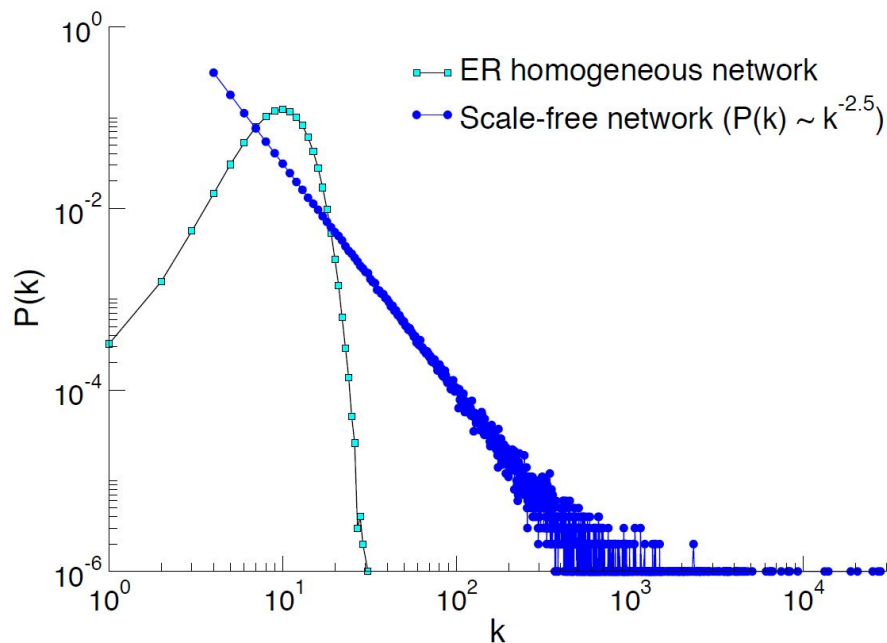
$E = \{\{v_1, v_2\}, \dots, \{v_n, v_m\}\} : \textit{Set of edges}$

$A = (a_{ij}) : \textit{Adjacency matrix},$

where $a_{ij} = \begin{cases} 0 & \textit{if } \{v_i, v_j\} \in E, \\ 1 & \textit{if } \{v_i, v_j\} \notin E \end{cases}$

03 - Graph analysis applied to resting-state fMRI

Random graphs, scale-free property and small wordness



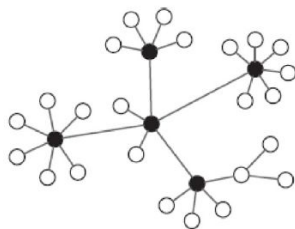
Boronchelli et al., 2013

	Random	Small-world	Scale-free
Degree distribution	Poisson	Peaked	Power law
Characteristic path length	Scales as $\log(N)$	Scales linearly with N for small p . And for higher p scales as $\log(N)$	Scales as $\log(N)/\log(\log N)$
Clustering coefficient	p (the connection probability)	High, but as $p \rightarrow 1$ behaves like a random graph	$((m-1)/2) * (\log(N)/N)$ where m is the number of edges with which a node enters
Robustness to failures	Similar responses to both random and targeted attacks	Similar response as random networks. This is because it has a degree distribution similar to random networks.	Highly resilient to random failures while being very sensitive to targeted attacks

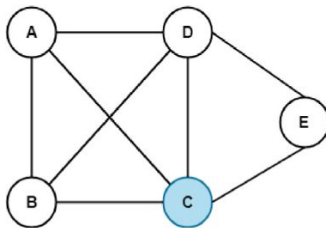
Thadakamalla et al., 2004

03 - Graph analysis applied to resting-state fMRI

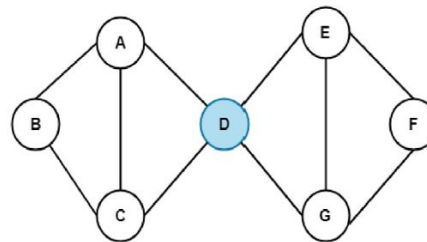
Centrality measures



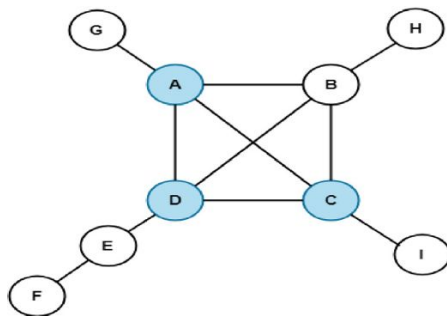
(A) Degree Centrality



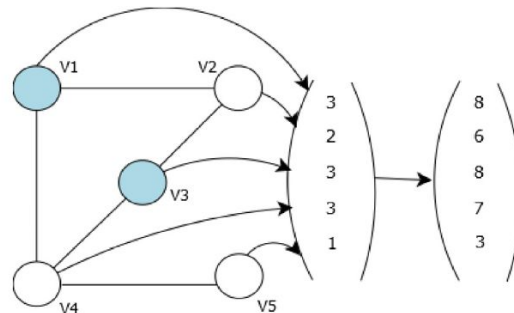
(B) Closeness Centrality



(C) Betweenness Centrality



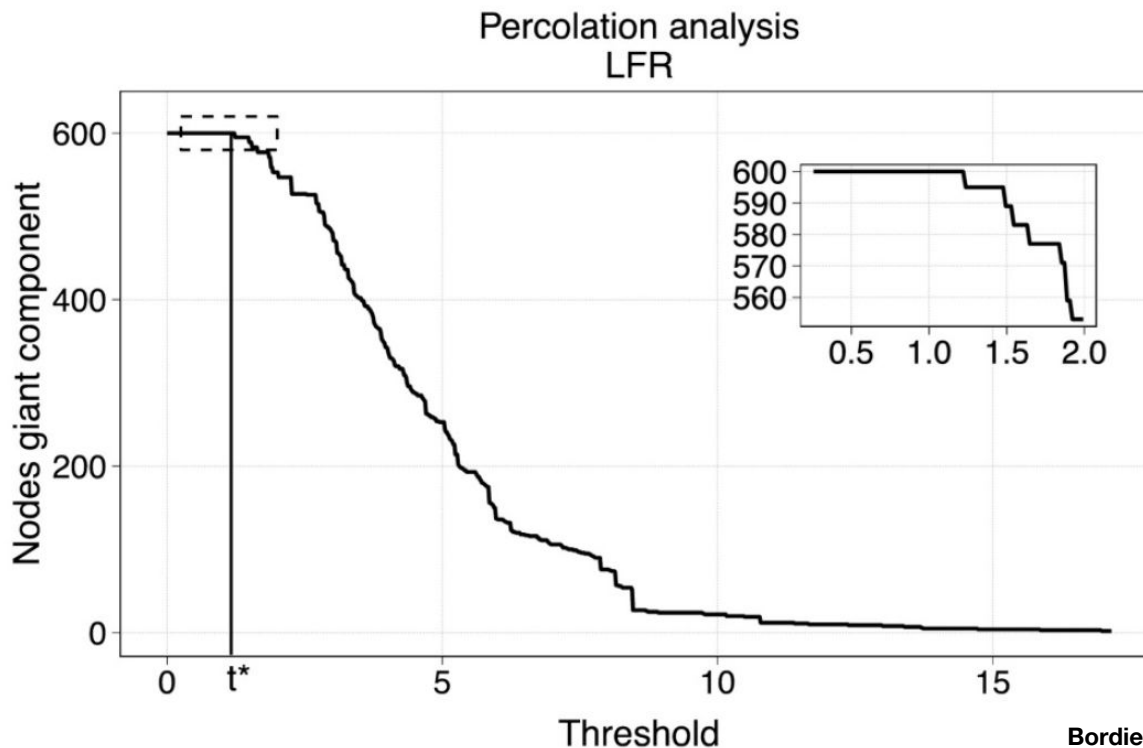
(D). Jordan Centrality



(E) Eigenvector Centrality

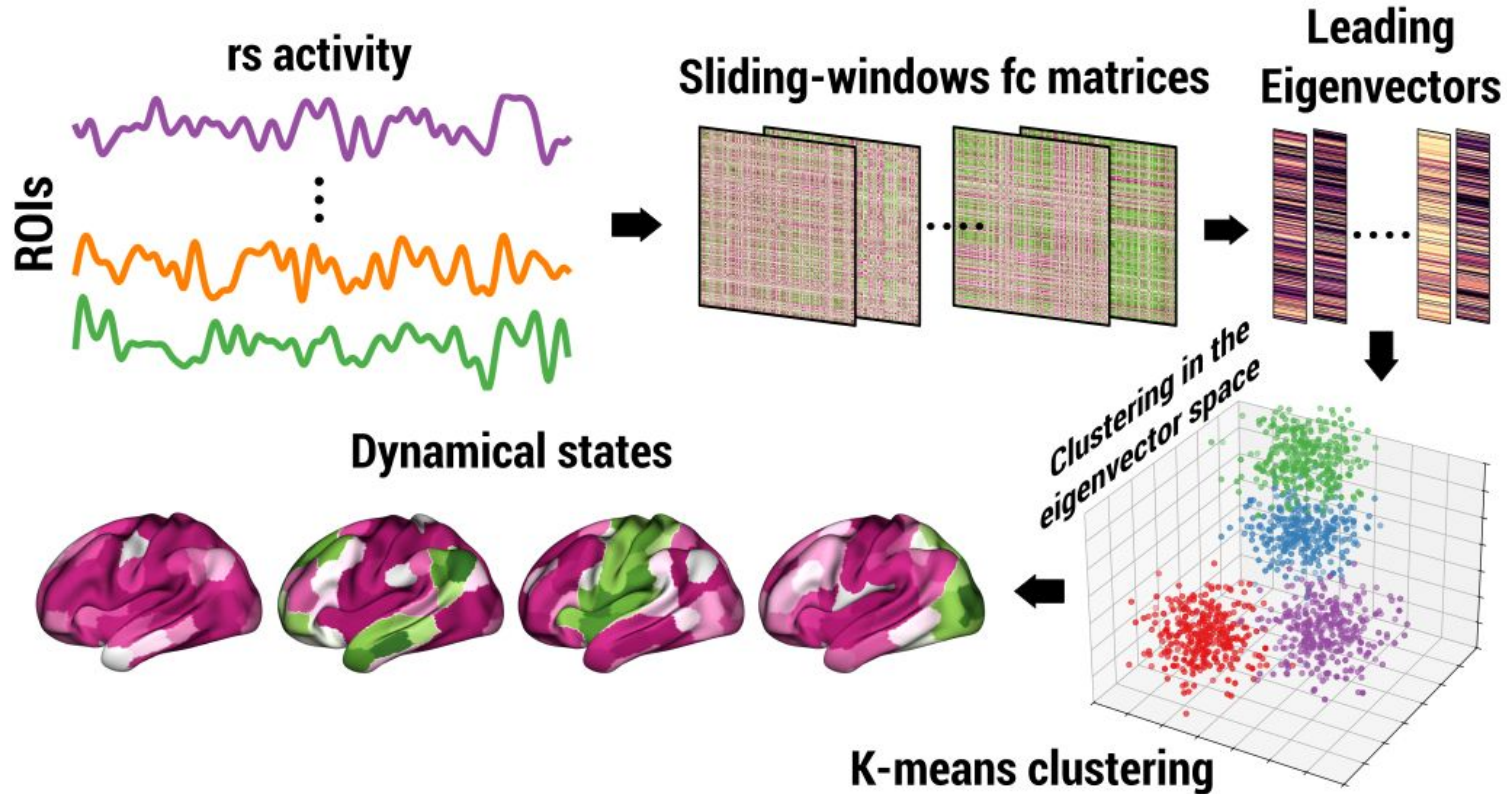
03 - Graph analysis applied to resting-state fMRI

Percolation



Bordier et al., 2013

04 - Dynamic Functional Connectivity



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